

Zulqernein Lakha

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SKILLS AND QUALIFICATIONS

- Looking to build company experience in Automotive Engineering with Business Management
- Proficient in C++, Arduino & PLC, AutoCAD, SolidWorks, Simulink, CATIA and MS Office
- Collaboration and leadership skills gained through teams and group projects
- Experienced in Database management and documentation
- Deadline compliant and reliable for management of tasks with different priorities
- Detail & results oriented with strong problem-solving skills and a strong desire to learn
- Flexible in fast-paced work environments

EDUCATION

BTECH, Automotive & Vehicle Engineering Technology, McMaster University, School of Engineering Practice & Technology, Hamilton, ON

- An integrated four-year joint engineering technology and accredited management program with McMaster University and Mohawk College

General Arts & Science Engineering preparatory program, Durham College

- Highest honors in program for year of enrollment.
- Preparatory studies for university qualification

High School Certificate, The City School International

- Completed IGCSE studies of eight courses with a mixture of entrepreneurship and science

Course Work

- **Mechatronics:** Integrating programming, electrical controls and mechanical systems using programmable logic controllers (PLC) and Arduino based microcontrollers (MCs)
- **Computer Aided Design:** In-depth study of design and intricate modelling of engine components and other various structural identities used in automobiles with the help of Catia & SolidWorks
- **Manufacturing Processes & Systems:** Study of various types of manufacturing methods like metal-casting, forming, shaping, injection moulding processes for metals, plastics, & ceramics
- **Structure & Properties of Materials:** Study of properties of metals, fracture and failure, phase diagrams, FMEA, engineering plastics and polymers, imperfections in solids and composites
- **Automotive Engineering Technology:** Understanding combustion cycles, timing, induction methods, sensors, and other performance and efficiency related aspects of vehicles
- **Design of Machine Elements:** Covers the theory and methodology related to conceptual design of machine elements used in automotive engineering including various forms of stress, fatigue, and in-depth analysis of fasteners, sliding and rolling bearings, gears, shafts, and brakes
- **Quality Control:** Study of statistical tools, tests, design and analysis of planned experiments, a further look into control charts for variables and attributes and ISO 9000 Certification
- **Vehicle Dynamics:** Learning Road vehicle dynamics including performance, handling, and ride of ground motor vehicles

Projects

- **Engineering Design – Hand Cart Model**
Designed and modeled a hand cart using SolidWorks to meet specifications, dimensions, and speed requirements in a team of four
- **Detailed modeling – Mazda Rotary Engine**
Replicating a rotary engine into a 3D CAD model with the help of CATIA using accurate dimensions and movement characteristics in a team of six
- **Capstone Project – Fire Fighting Robot**
Designed a robot tank with a fire extinguisher mount and microcontroller program as a 3D model on Autodesk Fusion 360, and further replicating the 3D model as a physical object using CNC Cut plastic tracks, Sheet metal Chassis, and Arduino programmed electrical components
- **Mechatronics Design – Colored cap sorting conveyor machine**
Built a conveyor belt run by a microcontroller that sorts colored bottle caps into the correct bins

Work Experience

Multimatic Inmet (February 2023 – April)

Production Operator, Richmond Hill, ON

- Checking fixtures and jigs to test & meet compliance of final product with client requirements
- Ensuring all production paperwork is in compliance with ISO 9001 & ISO 13485
- Follow control plans and work in accordance with company quality & safety specifications

Steel Art Signs Corporation (July 2022 – August 2022)

Machine Operator, Richmond Hill, ON

- Maintenance of machinery and equipment for daily use including monitoring for safety issues
- Sanding, drilling, along with metal fabrication using router machines and Brake Press
- MIG Welding and Tack Welding as per job orders

Ford Motor Company (January 2021 – September 2021)

Employee Safety Coach, Oakville Assembly Complex, ON

- Monitor body temperature of everyone in the facility using an infrared thermal imaging system
- Calibrating & diagnosing thermal imaging equipment and software across different departments

United Canada INC (April 2020 – September 2020)

Junior Operations Manager, Richmond Hill, ON

- Coordination of introducing new inventory for the company
- Prepare Warehouse inventory layouts using Just-In-Time and TQC methods for mask production
- Produce Verification & Validation documentation to ensure smooth processes in place

Hobbies & Activities

- All-rounder in sports, I enjoy ice skating, snorkeling, horse back riding, have also played table tennis for 15 years and Captain of championship winning High School basketball team
- Tutored Mathematics and Sciences along with other courses
- Love to take pictures of nature, passion for competitive PC Gaming and forest trail hiking